

Chapter 2: Water

Biochemistry

Generated by Cline

September 24, 2025

Outline

Weak Interactions in Aqueous Systems

Ionization of Water, Weak Acids, and Weak Bases

Buffering against pH Changes in Biological Systems

Outline

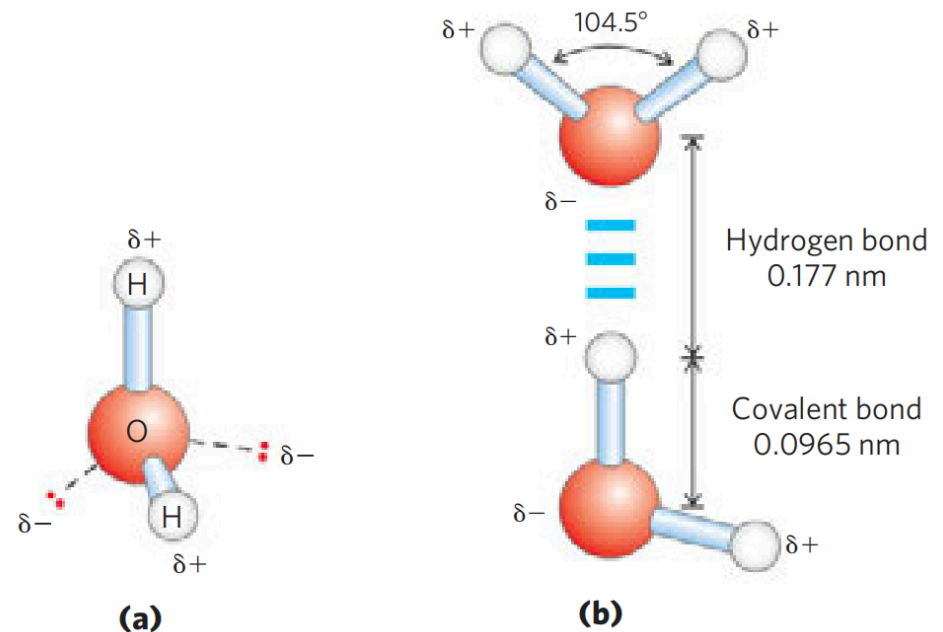
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Hydrogen Bonding Gives Water Its Unusual Properties

- Water constitutes 70% or more of the weight of most organisms.
- **Unusual Properties:** High melting point (0 °C), boiling point (100 °C), and heat of vaporization (2,260 J/g) due to strong intermolecular attractions. (See Table 2-1)
- **Molecular Structure:**
 - H₂O has a tetrahedral geometry with sp³ hybrid orbitals.
 - The H-O-H bond angle is 104.5°.

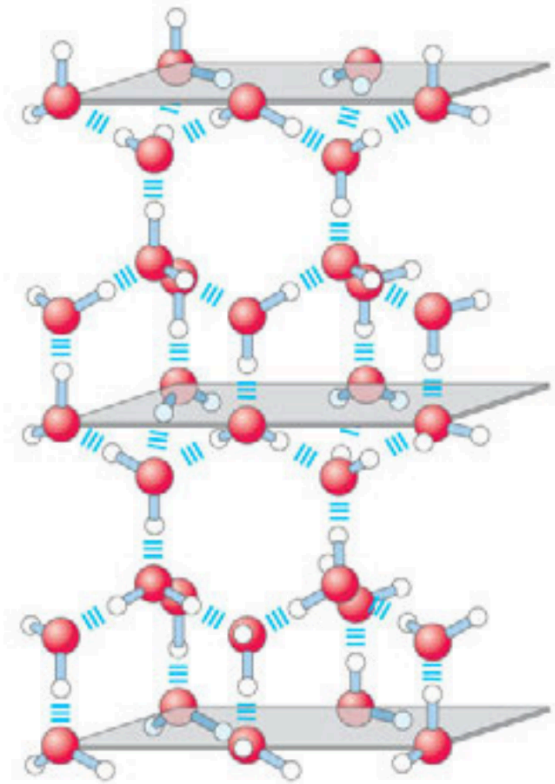


Hydrogen Bonding Gives Water Its Unusual Properties

- **Polarity:** Unequal electron sharing creates a permanent dipole ($2\delta^-$ on O, δ^+ on H).

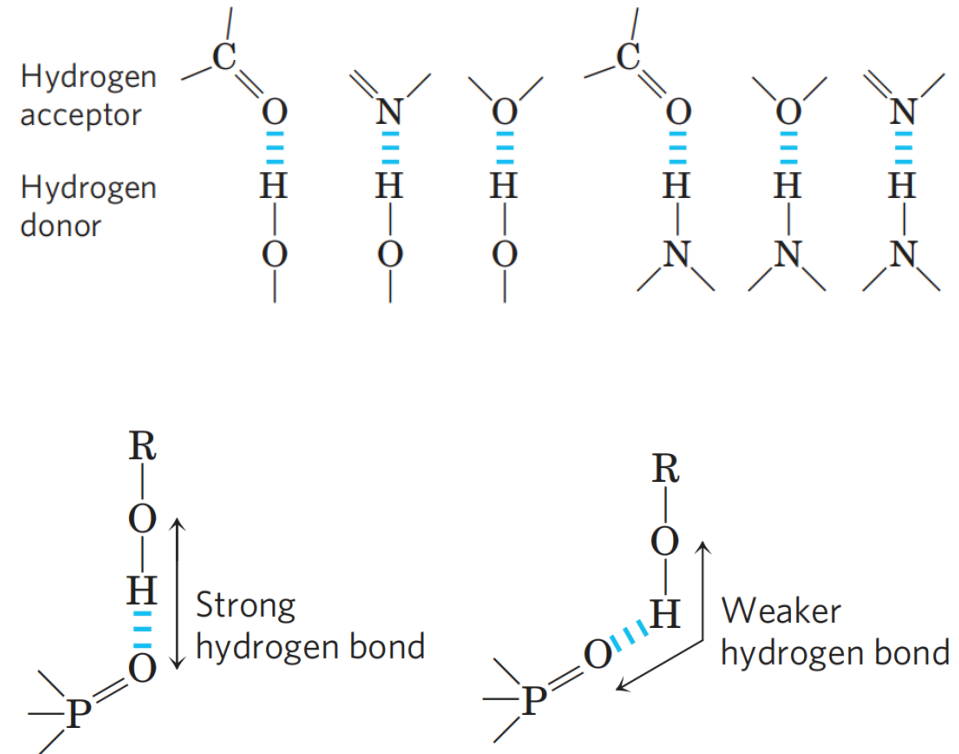
Hydrogen Bonding and Structure of Ice

- **Hydrogen Bond**: Electrostatic attraction between water molecules (23 kJ/mol).
- **Flickering Clusters**: In liquid water, H-bonds are transient (1-20 ps), with each molecule bonding to 3.4 others.
- **Structure of Ice**: Each water molecule forms four stable H-bonds, creating a regular crystal lattice.
- This ordered lattice makes ice less dense than liquid water.



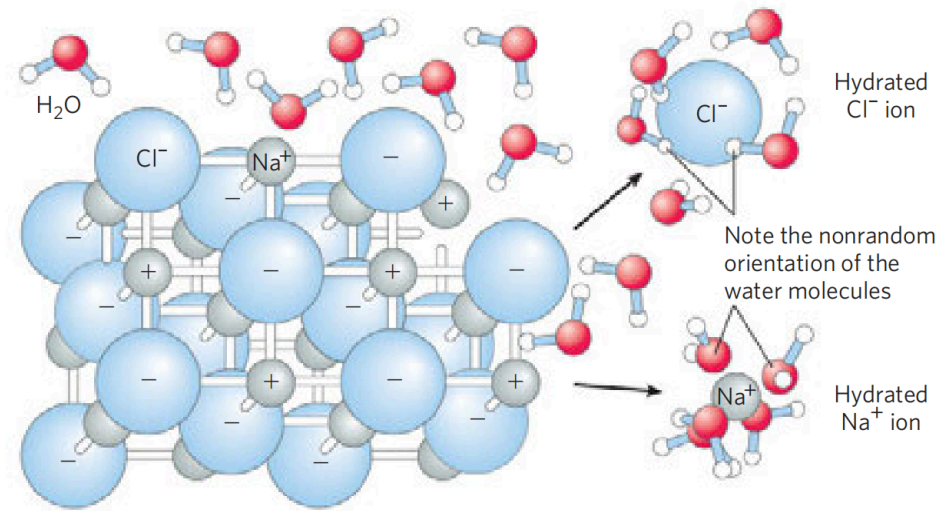
Water Forms Hydrogen Bonds with Polar Solutes

- Hydrogen bonds form between a hydrogen donor (H on an electronegative atom) and a hydrogen acceptor (O or N).
- Polar biomolecules (sugars, alcohols, etc.) dissolve in water by forming H-bonds.
- Directionality**: H-bonds are strongest when the donor, hydrogen, and acceptor are collinear, which is critical for macromolecular structure.



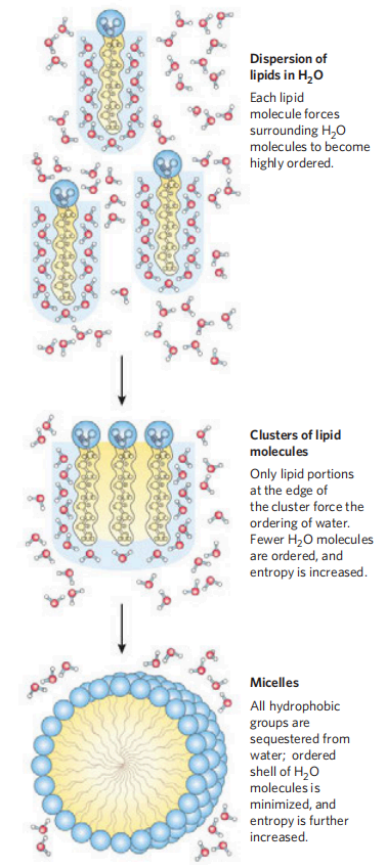
Water Interacts Electrostatically with Charged Solutes

- Water is a polar solvent for hydrophilic (charged or polar) compounds.
- It dissolves salts (e.g., NaCl) by hydrating and screening the ions (Na^+ , Cl^-).
- **High Dielectric Constant** ($\epsilon = 78.5$): Water weakens electrostatic interactions between dissolved ions, described by $F = \frac{Q_1 Q_2}{\epsilon r^2}$.



The Hydrophobic Effect

- **Nonpolar Compounds:** Poorly soluble in water as they force an ordered, entropically unfavorable “cage” of water molecules around them.
- **Hydrophobic Effect:** The tendency of nonpolar substances to aggregate in aqueous solution to minimize contact with water.
- **Amphipathic Compounds:** Contain both polar (hydrophilic) and nonpolar (hydrophobic) regions.

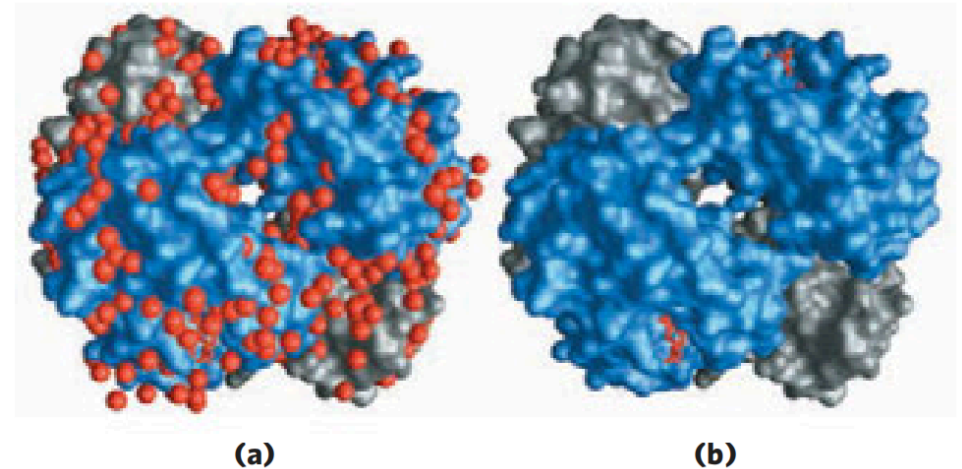


The Hydrophobic Effect

- In water, they form **micelles**, with hydrophobic tails hidden inside and hydrophilic heads facing outwards. This process is driven by the increase in entropy of the surrounding water.

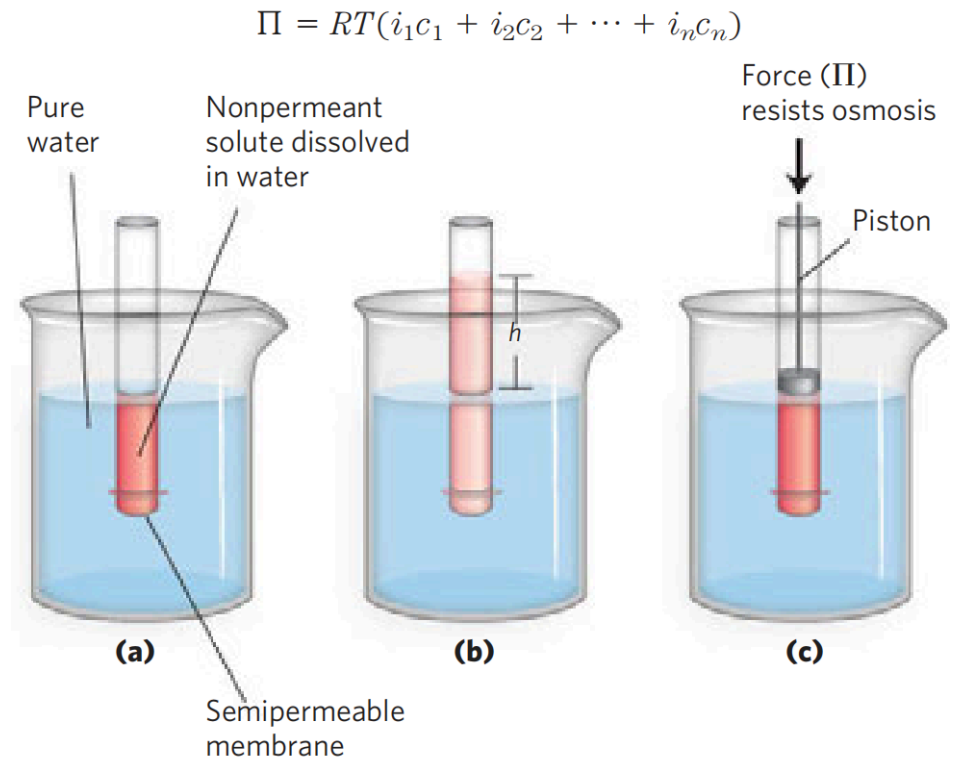
Weak Interactions in Biological Systems

- Four major types of noncovalent (“weak”) interactions:
 1. Hydrogen Bonds
 2. Ionic Interactions
 3. Hydrophobic Interactions
 4. van der Waals Interactions
- **Bound Water:** Water molecules can be tightly bound to macromolecules and are often essential for their function.



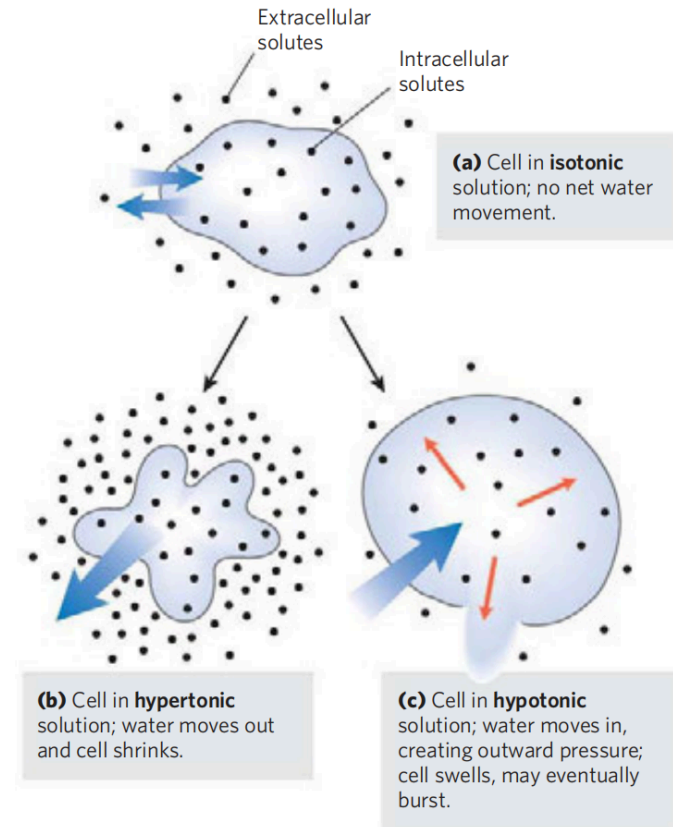
Colligative Properties and Osmosis

- **Colligative Properties:** Depend on the number of solute particles, not their identity (e.g., boiling point, freezing point).
- **Osmosis:** The movement of water across a semipermeable membrane from a region of high water concentration to low water concentration.
- **Osmotic Pressure** (Ψ / Π): The pressure required to stop osmosis, given by the van't Hoff equation: $\Pi = icRT$.



Tonicity and Cells

- **Isotonic:** Solution osmolarity equals cell's cytosol; no net water movement.
- **Hypertonic:** Solution osmolarity is higher; water moves out, and the cell shrinks.
- **Hypotonic:** Solution osmolarity is lower; water moves in, and the cell swells and may burst (lysis).
- Cells must maintain osmotic balance to survive.



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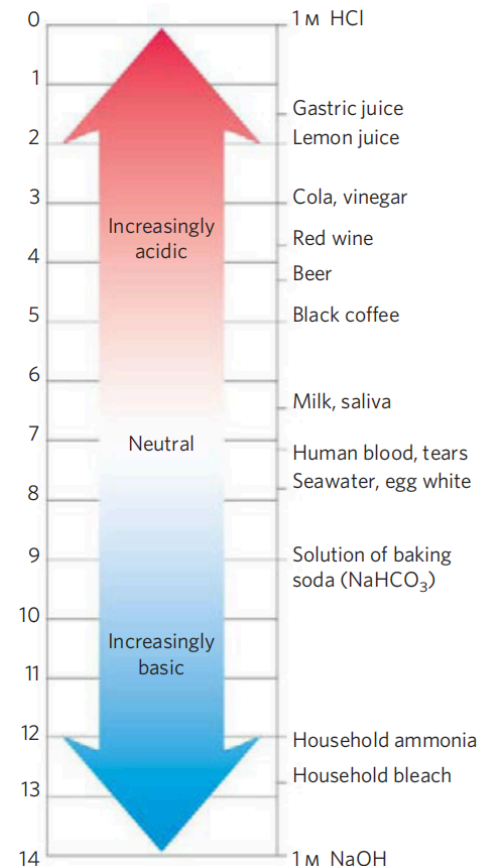
Weak Interactions in Aqueous Systems

Ionization of Water, Weak Acids, and Weak Bases

Buffering against pH Changes in Biological Systems

Ionization of Water and the pH Scale

- Water autoionizes: $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$.
- Protons exist as hydronium ions (H_3O^+).
- Ion Product of Water** (K_w): $K_w = [\text{H}^+][\text{OH}^-] = 1.0 \times 10^{-14} \text{ M}^2$ at 25 °C.
- pH Scale**: A logarithmic scale for $[\text{H}^+]$.
 - $\text{pH} = -\log[\text{H}^+]$
 - pH 7 is neutral.
 - pH < 7 is acidic.
 - pH > 7 is basic.



Weak Acids, Weak Bases, and pK_a

- **Weak acids** and **bases** only partially dissociate in water.
- A **conjugate acid-base pair** consists of a proton donor and its corresponding proton acceptor.
- **Acid Dissociation Constant** (K_a): Measures the strength of an acid. $K_a = \frac{[H^+][A^-]}{[HA]}$.
- **pK_a** : The negative logarithm of K_a .
 - $pK_a = -\log(K_a)$
 - A smaller pK_a indicates a stronger acid.

Weak Acids, Weak Bases, and pK_a

Monoprotic acids

Acetic acid
($K_a = 1.74 \times 10^{-5} \text{ M}$)

Ammonium ion
($K_a = 5.62 \times 10^{-10} \text{ M}$)

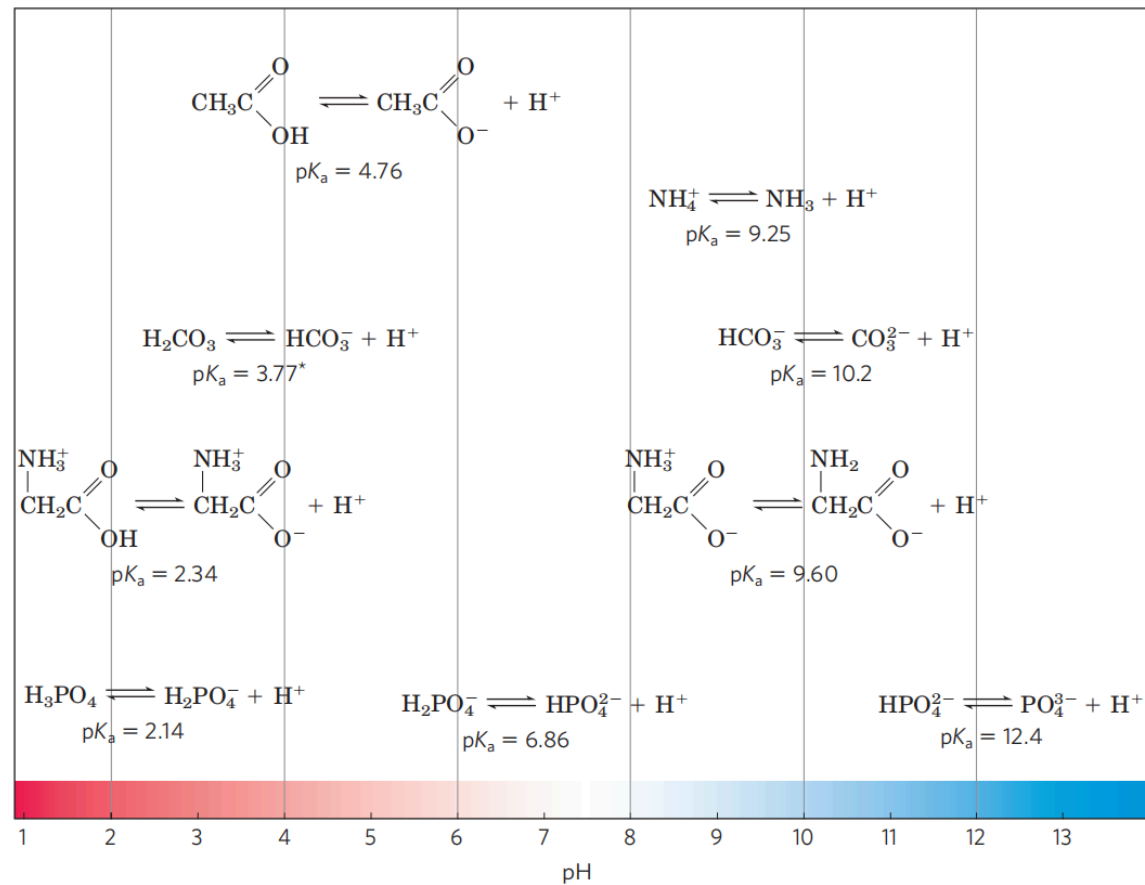
Diprotic acids

Carbonic acid
($K_a = 1.70 \times 10^{-4} \text{ M}$);
Bicarbonate
($K_a = 6.31 \times 10^{-11} \text{ M}$)

Glycine, carboxyl
($K_a = 4.57 \times 10^{-3} \text{ M}$);
Glycine, amino
($K_a = 2.51 \times 10^{-10} \text{ M}$)

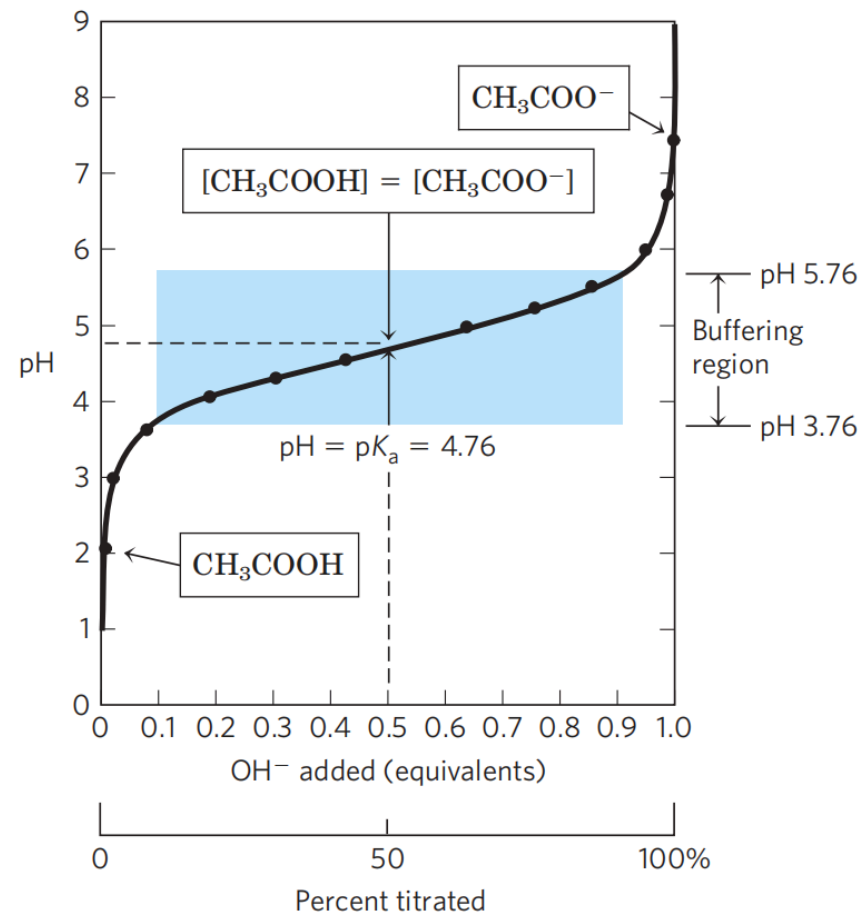
Triprotic acids

Phosphoric acid
($K_a = 7.25 \times 10^{-3} \text{ M}$);
Dihydrogen phosphate
($K_a = 1.38 \times 10^{-7} \text{ M}$);
Monohydrogen phosphate
($K_a = 3.98 \times 10^{-13} \text{ M}$)



Titration Curves

- A **titration curve** plots pH against the amount of base added to an acid.
- The flat region of the curve is the **buffering region**.
- At the midpoint of the titration, $[\text{HA}] = [\text{A}^-]$, and the pH is numerically equal to the $\text{p}K_a$.



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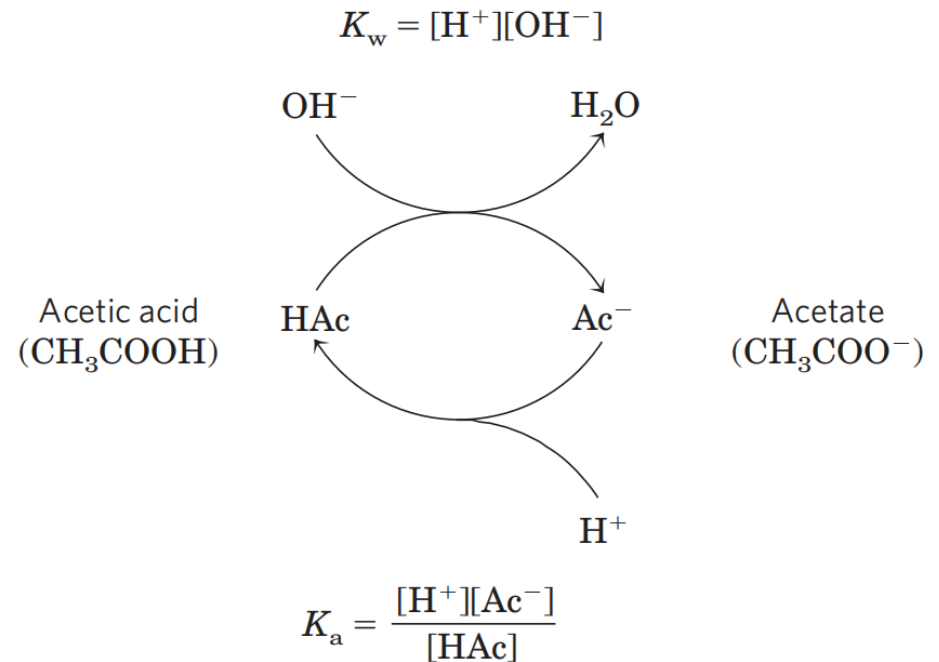
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Buffers and the Henderson-Hasselbalch Equation

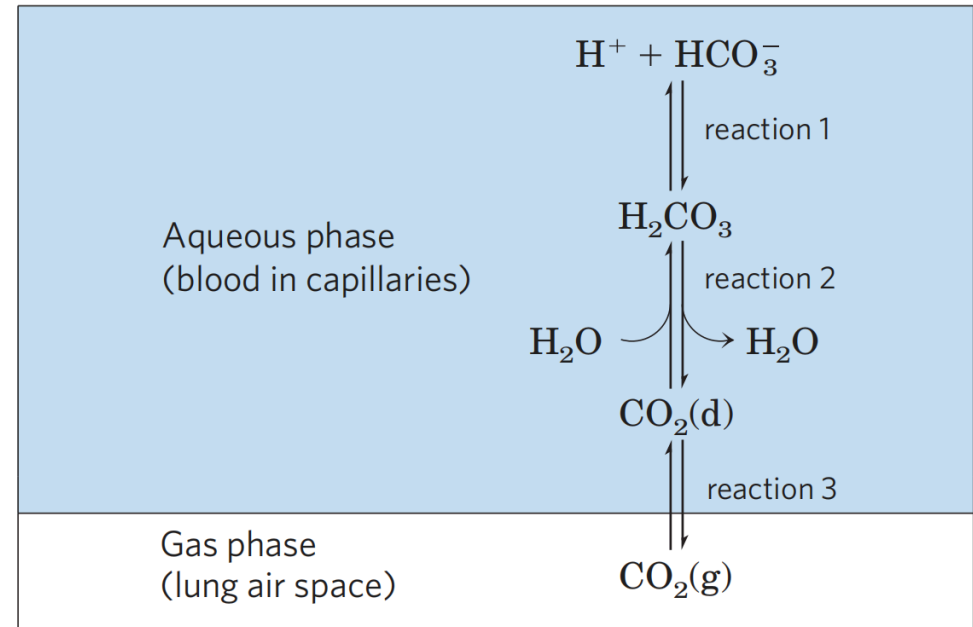
- A **buffer** is a mixture of a weak acid and its conjugate base that resists pH changes.
- Buffering capacity is maximal at the pK_a and effective within $pK_a \pm 1$ pH unit.
- **Henderson-Hasselbalch Equation:**

$$\text{pH} = \text{p}K_a + \log\left(\frac{[\text{A}^-]}{[\text{HA}]}\right)$$
- This equation relates pH, pK_a , and the buffer component ratio.

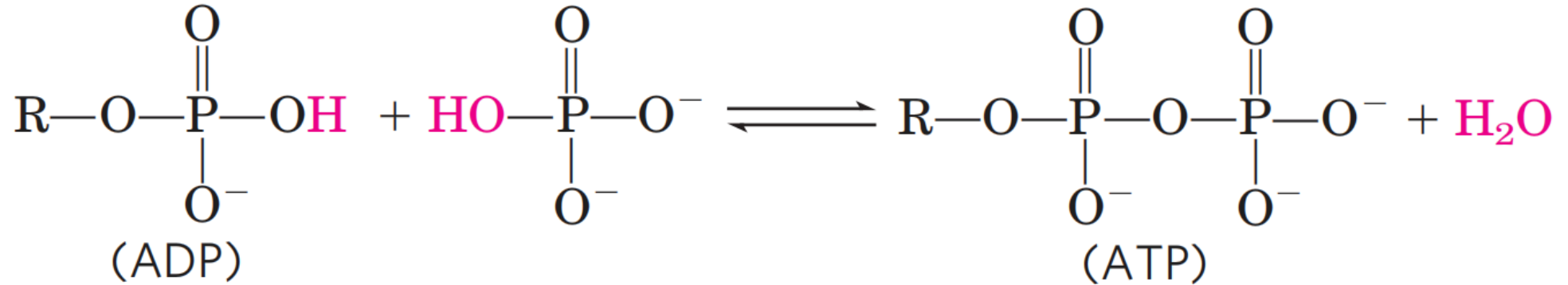


Biological Buffer Systems

- Biological fluids are buffered to maintain a constant pH.
- **Phosphate Buffer System:**
 - Main intracellular buffer.
 - Pair: $\text{H}_2\text{PO}_4^- / \text{HPO}_4^{2-}$
 - $\text{p}K_a = 6.86$, effective near neutral pH.
- **Bicarbonate Buffer System:**
 - Main blood plasma buffer.
 - Equilibrium: $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3 \rightleftharpoons \text{H}^+ + \text{HCO}_3^-$



Water as a Reactant



Phosphoanhydride

- Water is a direct participant in many metabolic reactions.
- **Hydrolysis**: Cleavage of a bond by adding water.
- **Condensation**: Formation of a bond by removing water.
- Example: ATP is formed from ADP and phosphate via condensation.